Group X Progress Report:
My Group's Project Name

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1 Introduction

Pneumonia is a severe respiratory infection causing significant morbidity and mortality worldwide, especially in young children. Accurate and timely diagnosis is crucial for effective treatment and improving outcomes. Traditionally, diagnosis relies on expert radiologists interpreting chest X-rays, but deep learning-based automated detection systems offer potential for faster and scalable assessment. Our project aims to develop a binary classification model that distinguishes between normal and bacterial pneumonia pediatric chest X-ray images using convolutional neural networks (CNNs) and transfer learning with pretrained models.

2 Related Work

3 Dataset

We use the publicly available pediatric chest X-ray dataset sourced from Guangzhou Women and Children's Medical Center (**?**). It contains labeled X-ray images classified as Normal, Viral Pneumonia, and Bacterial Pneumonia. For this project, we focus on binary classification of Normal vs. Bacterial Pneumonia images. The dataset is split stratified into 80% training, 10% validation, and 10% test sets.

Preprocessing includes resizing all images to 224×224 pixels, normalizing pixel intensities to $[0, 1]$, and data augmentation with random cropping, horizontal flipping, and slight rotation.

4 Features

Our approach uses the CNN to automatically learn hierarchical feature representations from input X-ray images. The network inputs are augmented and normalized images, optionally processed with histogram equalization to improve contrast. No manual feature engineering is performed beyond preprocessing. Data representations are learned end-to-end by fine-tuning the pretrained ResNet18 model on the dataset.

5 Implementation

The model implementation centers on transfer learning with a ResNet18 backbone pretrained on ImageNet. Initially, only the classification head is trained while freezing earlier layers ("head-only" training phase), followed by fine-tuning deeper convolutional layers ("unfreezing last block"). Training uses weighted cross-entropy loss to address class imbalance, the optimizer, learning rate modification , and early stopping based on validation F1 score.

6 Results and Evaluation

We evaluate model performance using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices on validation and test sets. The majority-class baseline serves as a simple comparative benchmark. Preliminary results show that the CNN model consistently outperforms the baseline, achieving higher recall and F1 scores especially critical for pneumonia detection.

7 Feedback and Plans

Write about your plans for the remainder of the project. This should include a discussion of the feedback you received from your TA, and how you plan to improve your approach. Reflect on your implementation and areas for improvement. Refer to item 6. This may be around 0.5 pages.

8 Template Notes

You can remove this section or comment it out, as it only contains instructions for how to use this template. You may use subsections in your document as you find appropriate.

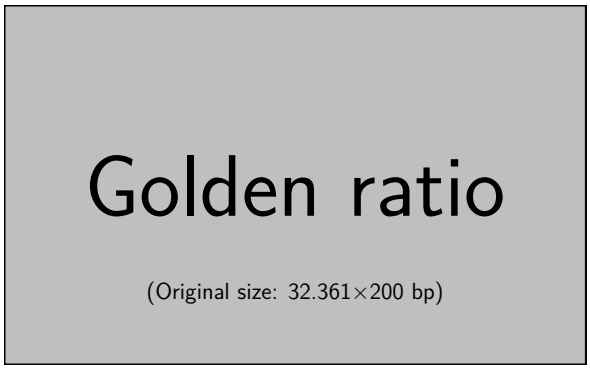


Figure 1: A figure with a caption that runs for more than one line. Example image is usually available through the `mwe` package without even mentioning it in the preamble.

8.1 Tables and figures

See Table 1 for an example of a table and its caption. See Figure 1 for an example of a figure and its caption.

8.2 Citations

Table 1 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get "author (year)" citations, like this citation to a paper by Gusfield (1997). You can use the command `\citep` (cite in parentheses) to get "(author, year)" citations (Gusfield, 1997). You can use the command `\citealp` (alternative cite without parentheses) to get "author, year" citations, which is useful for using citations within parentheses (e.g. Gusfield, 1997).

8.3 References

Many websites where you can find academic papers also allow you to export a bib file for citation or bib formatted entry. Copy this into the `custom.bib` and you will be able to cite the paper in the LaTeX. You can remove the example entries.

8.4 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command. This an example cross-reference to Equation 1. You can also write equations inline, like this: $A = \pi r^2$.

Team Contributions

Write in this section a few sentences describing the contributions of each team member. What did each member work on? Refer to item 7.

References

Rie Kubota Ando and Tong Zhang. 2005. A framework for learning predictive structures from multiple tasks and unlabeled data. *Journal of Machine Learning Research*, 6:1817–1853.

Galen Andrew and Jianfeng Gao. 2007. Scalable training of L1-regularized log-linear models. In *Proceedings of the 24th International Conference on Machine Learning*, pages 33–40.

Dan Gusfield. 1997. *Algorithms on Strings, Trees and Sequences*. Cambridge University Press, Cambridge, UK.

Mohammad Sadegh Rasooli and Joel R. Tetreault. 2015. Yara parser: A fast and accurate dependency parser. *Computing Research Repository*, arXiv:1503.06733. Version 2.

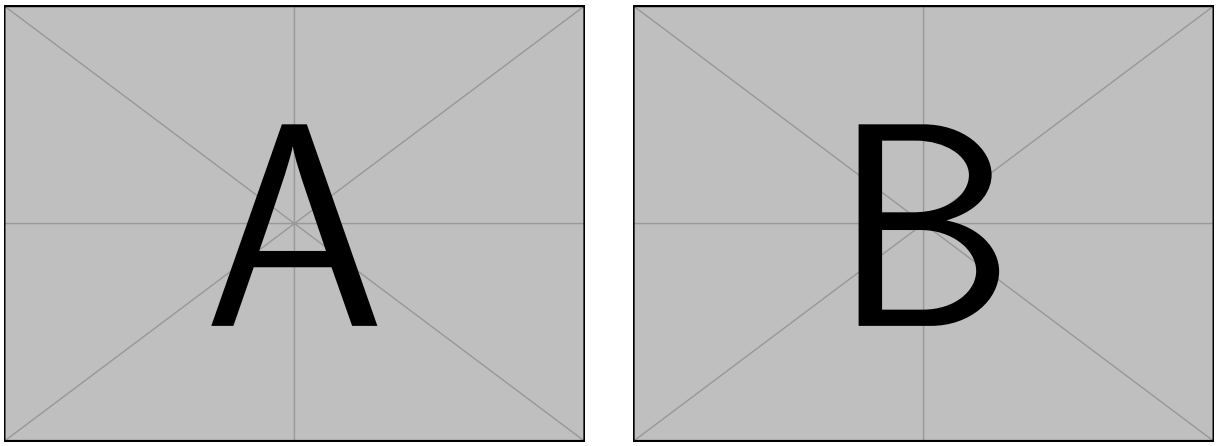


Figure 2: A minimal working example to demonstrate how to place two images side-by-side.

Output	natbib command	ACL only command
(Gusfield, 1997)	`\citep`	
Gusfield, 1997	`\citealp`	
Gusfield (1997)	`\citet`	
(1997)	`\citeyearpar`	
Gusfield's (1997)		`\citeposs`

Table 1: Citation commands supported by the style file.